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Optimizing Shale Assets: A Hybrid Approach Combining Classification and Prescriptive Analytics to Boost Well Performance in the Eagle Ford Basin

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Abstract

This Project seeks to explain the characteristics and variables that differentiate good wells from bad wells in the Eagle Ford Basin using Net Present Value (NPV) as a classifier. The first part of the project comprises of building a robust classifier that can differentiate good wells from bad wells. The second part of the project is the development of different classification models, variable selection and feature engineering, and the selection of the best performing model. The third part of the project entails identifying features that impact NPV. Lastly, the model is used to characterized what engineering variables best improve the NPV of deficient wells.

The dataset was cleaned and processed for missing values and outliers. The variables in the dataset were divided into 3 main classes. The first class is the target or dependent variable which is the NPV. Geological independent variables which cannot be controlled or engineered is the second class of variables, and engineering independent variables which can be controlled is the last. A classifier was created using a simple rule and tested to be balanced. Different classification algorithms, Generative, Discriminative and Ensemble was applied to the dataset and the model with the best classification score was selected. Prescriptive analytics was carried out with the selected model on the wells with negative NPV to determine the optimum engineering variables that can make these wells positive.

Random Forest, an Ensemble model, outperformed all the other models. For most of the models, the location of the well and TVD (geological variables), the proppant volume & lateral length (engineering variables) strongly influences well NPV classification. This methodology serves as a quick way for oil and gas operators to strongly identify variables that affect NPV and also to select the optimum engineering values which can complement other established methods such as Linear Regression Analysis, Rate Transient Analysis or Reservoir Simulation on a fairly large dataset.

INTRODUCTION

Multiple variables affect Oil and Gas Estimated Ultimate Recovery (EUR) in unconventional reservoirs. These unconventional wells must be stimulated (hydraulically fractured) to be economically variable. The intensity of the stimulation should definitely affect the level of output we obtain from these wells. Components that go into stimulating a hydraulically fractured well include cluster spacing, stages per well,

amount of proppant (sand), and amount of fluid (water). Better output can be achieved by increasing the intensity of the stimulation, but this comes at a cost. Therefore, there is a practical limit to how much we can raise the well's production. These well stimulation variables can be controlled or engineered.

In addition to well stimulation variables, there is a separate collection of factors known as well-engineering variables. These include well spacing, and well lateral length. These variables can also be controlled or engineered. In this paper, we refer to both the well stimulation and well engineering variables as Engineering variables.

Another set of variables that affect oil and gas production in unconventional reservoir is petrophysical and geomechanical variables. These variables include porosity, permeability, water saturation, total organic carbon content, minimum horizontal stress, Poisson's ratio, and Young's modulus. It is impossible to regulate or design these variables. The engineering variables, geological variables and independent variable (target variable) are summarized in [table 1](#). Below are explanations of each feature

- Lat: The x coordinate of the well (continuous)
- Long: The y coordinate of the well (continuous)
- TVD: The depth of the well (continuous)
- Oil gravity: The type of oil that was produced (4 classes: heavy condensate, light, light condensate, ultra-light) (discrete)
- Prop_Vol: The amount of sand that is used to keep the frac opened (continuous)
- Fluid_Vol: The amount of water that pumped to carry the sand (continuous)
- LL: The horizontal length of the well (continuous)
- Well spacing: The distance from one well to another well (continuous)
- NPV: Net present value of the well (continuous)

Table 1—Features that control oil and gas production

Geological Variables (uncontrollable)	Engineering Variables (Controllable)	Target Variable
Lat (latitude)	Prop_vol (Amount proppant)	NPV
Long (longitude)	Fluid_vol (Amount of fluid)	
True vertical depth (TVD)	LL (Lateral Length)	
Oil gravity (Weight of Oil)	Well Spacing	

Simple base models like the decline curve analysis (production data fitted to a model with hyperbolic or exponential parameters) can be used to estimate EUR. These models cannot explain what features are responsible for production because all the features have been parameterized into a single exponent. Moreover, ([Okouma et al., 2012](#)), suggest that Arps-type models often fail in low-permeability environments due to invalid assumptions like boundary-dominated flow, constant bottomhole pressure.

Physics based models can be used to estimate the EUR of these wells. And in theory, these models should be able to explain what features result in a good or bad EUR. However, developing such models can be complex and time-consuming. Moreover, they can only be used by trained professionals who have access to expensive software.

In this work, NPV (net present value) is used to calculate the profitability of a well. A simple formulation for the calculation of NPV is below:

$$NPV = EUR * Price - Wellcost$$

The Estimated Ultimate Recovery (EUR) is a critical component of this calculation, representing the total amount of oil or gas expected to be recovered from a well. This value is usually expressed in units such as barrels of oil (bbl) or million cubic feet of gas (MMcf). The Price of oil or gas per unit is also a key factor, as it determines the revenue generated by the well. Where the well cost is the cost of drilling the well (well engineering variables), and the cost of stimulating the well.

Since oil companies do not disclose petrophysical and geomechanical properties of wells publicly, therefore these properties will be proxied using well location latitude and longitude under the assumption that spatial distribution of the geomechanical and petrophysical properties should be roughly uniform across a given region.

With the proliferation of drilled well in the Eagle Ford Basin, EUR estimates can quickly be generated using simple based model like decline curve analysis, and NPV calculated afterwards. This paper proposes a machine-learning model that can:

- Classify whether a well will give a good NPV or a bad NPV.
- Identify the factors that most define why certain wells have favorable NPVs while others do not.
- Prescribe what adjustments can be changed to make a bad well better.

DATA PROCESSING

The data was collected from the IHS Markit website. [Figure 1](#) shows that over 30,000 wells were collected from the website. To ensure reliable EUR estimates, only wells with 365 days of production was selected for the analysis. The 365-day values (BOE) range from min, red (value of 1) to max blue (value of 1,229,020) with the median value being white. The value represents the amount of oil and gas that was extracted in the first 365 days. The black color represents wells that have no values. Data with no value or with a lateral length (LL) of less than 4,000 ft and greater than 12,000 ft were also discarded.

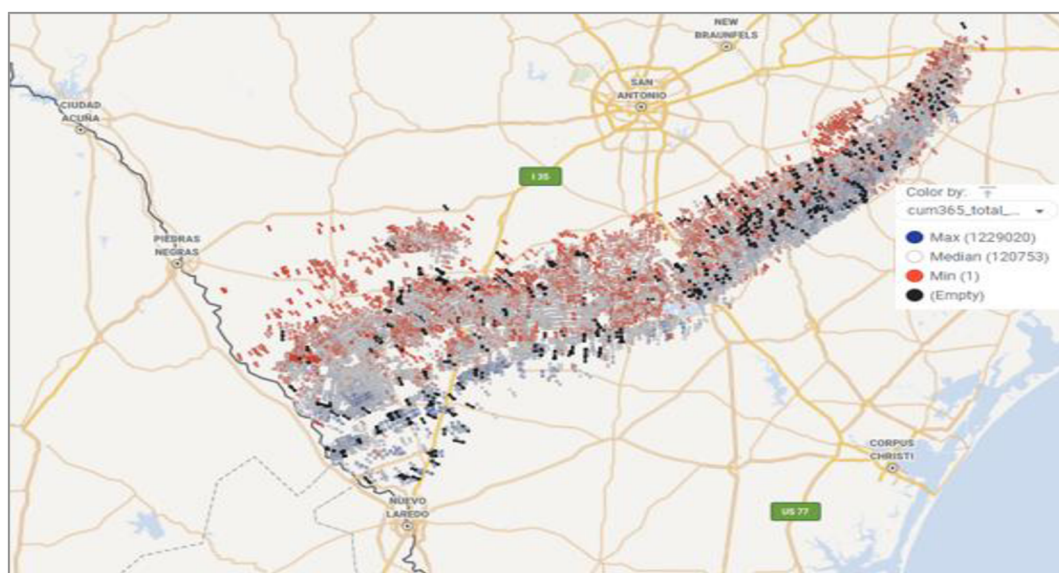


Figure 1—Location of well. Color by production

Figure 2 shows that after the data processing, only 4,600 wells remained. The color is coded by NPV where the red represents the min NPV (- \$ 21 M) whereas the blue represents the max NPV (\$86M).

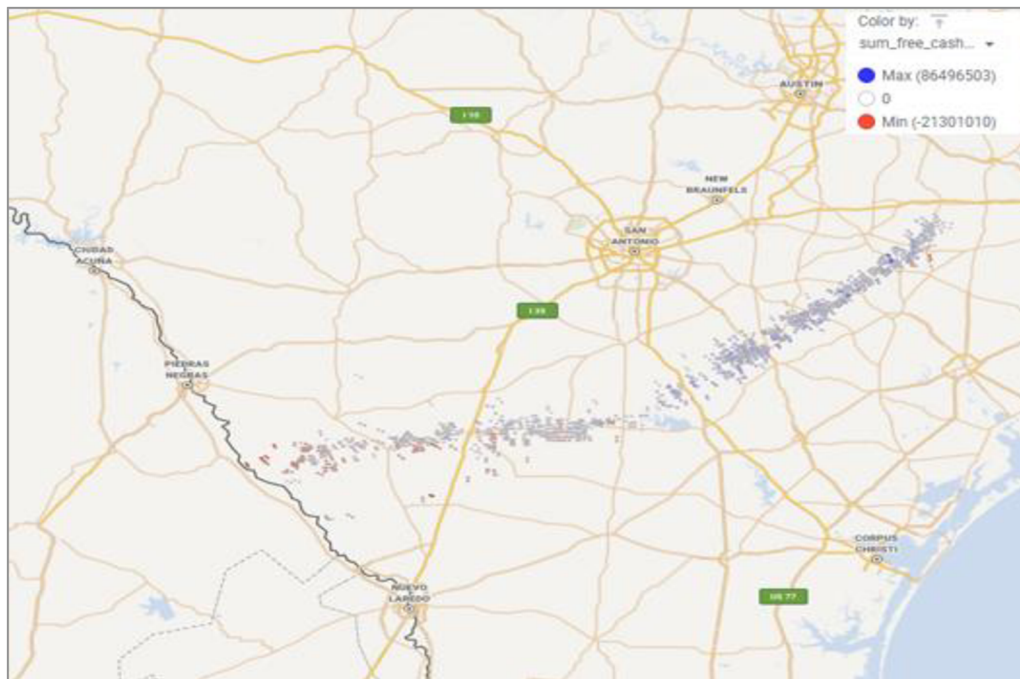


Figure 2—Location of wells after cleaning. Color by NPV

METHODOLOGY

Making the classifier.

First, a classifier was created from the Target Variable (NPV) using the following rule.

$$NPV01 = 0, NPV < 0$$

$$NPV01 = 1, NPV > 0$$

Where NPV01 is the classifier.

A common problem in classification algorithms is class imbalance. When observation in one class is higher than in other classes, a class imbalance exists. This can severely affect the result of the algorithm (Wei and Dunbrack Jr, 2013), suggest that a 50-50 distribution of both classes results in the highest balanced accuracy.

A count of the wells with negative NPV is 1653 while that of the positive NPV is 2954. The calculation of the class proportion is revealed to be 0.64 for the positive NPV wells and 0.36 for the wells with negative NPV. (Brownlee, 2020), from machine learning mastery note that 60/40 ratio is often considered a mildly imbalanced and is often not a concern, while (Buda et al., 2017), indicates that imbalance 100/1 ratio is considered extreme in classification tasks. Lastly a visual check of Fig. 3 shows that no one class dominates. This reveals there is no severe imbalance in the dataset.

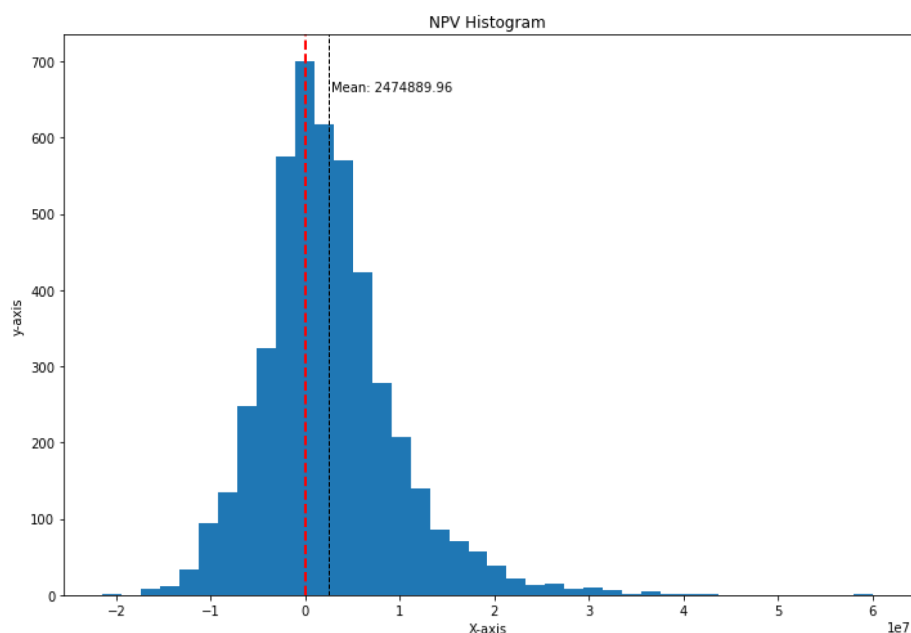


Figure 3—Histogram of the NPV shows that the classifier (Zero NPV) in red and the mean in black.

Exploratory Data Analysis

Exploratory Data Analysis was performed to see what features correlate with the target variable (NPV). The correlation matrix in Figure 4 describes the relationship between all the variables. Any features with an r -value of equal or greater than 0.3 were interpreted to correlate with the NPV. These features include:

1. LL
2. Prop Vol
3. Fluid Vol
4. TVD
5. Lat
6. Long

The geological variables seem to correlate more strongly with NPV than the engineering variables. After this, 13 different models were used to classify the target variable NPV01 into two classes. Class 1 is a dataset less than zero, while Class 2 is a dataset greater than zero.

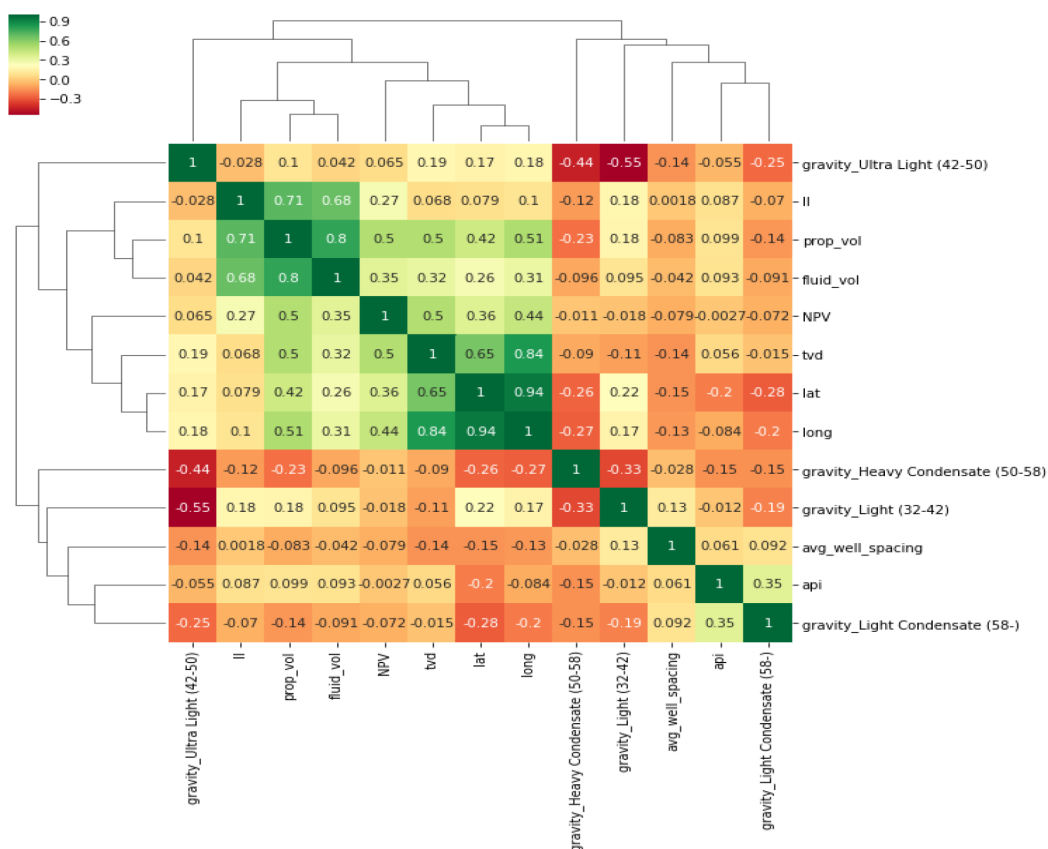


Figure 4—Correlation Matrix

Models Description

Table 2 below summarizes the 13 different model types. Rather than using AIC or BIC methods to do the feature selection, a thorough search was conducted to identify the set of features that produces the best classification result. The exhaustive search method generates 2^p models, where p is the number of features, and the model with the best classification was used to determine what features best distinguish the two classes in the target variable from one another. For instance, for the exhaustive search method for model 2, 2^{11} (2048) sub models were generated and the combination of features that gives the best score was selected. For the ensemble methods, feature selection was performed based on the mean decrease in impurity. A brief description of each model is given below.

Table 2—Practical Ranges from imbalance classes for classification problems. From (Brownlee, 2020; Buda et al., 2017).

Imbalance Type	Ratio Example	Notes
Balanced	~1:1 (50%/50%)	Ideal condition
Mild imbalance	~1.5:1 (60/40)	Usually, tolerable
Moderate imbalance	~4:1 (80/20)	May need adjustments and resampling
Severe imbalance	≥100:1	Demands advanced handling: oversampling, cost-sensitive learning, threshold tuning
Extreme imbalance	≥1000:1	For rare events (fraud, medical), requires specialized methods

Table 3—Summary of the utilized models

Model Family	Model
Generative	Model 1 (LDA, with all features)
	Model 2 (LDA, exhaustive search)
	Model 3 (QDA, with all features)
	Model 4 (QDA, exhaustive search)
	Model 5 (NB, with all features)
	Model 6 (NB, exhaustive search)
Discriminative	Model 7 (KNN all features)
	Model 8 (KNN, exhaustive search)
	Model 9 (PCA then KNN)
Ensemble	Model 10 (Random forest with default parameters)
	Model 11 (Random forest with tuned parameters)
	Model 12 (Gradient Boost with default parameters)
	Model 13 (Gradient Boost with tuned parameters)

Further description of the models can be found in The Elements of Statistical Learning².

Model 1 &2: Enhanced LDA with Exhaustive Search and Feature Selection

Model 2 builds upon the foundation of Model 1, which utilizes Linear Discriminant Analysis (LDA) to distinguish between two or more classes using a linear combination of features. However, Model 2 addresses some of the limitations of Model 1 by incorporating an exhaustive search methodology for feature selection. This approach generates 2^p models (where p is the number of features) and selects the model with the best combination of features that yields the highest f1-score.

While Model 2 inherits some characteristics from Model 1, such as:

- Requiring all features to follow a normal distribution
- Requiring covariances to be the same for all classes
- Not necessitating scaling

It also improves upon Model 1 in several ways:

1. Handling collinearity: By using an exhaustive search for feature selection, Model 2 can better handle collinearity issues, which were a concern in Model 1.
2. Variable selection: The exhaustive search approach allows for more robust variable selection, as it evaluates all possible combinations of features to find the optimal subset.
3. Performance evaluation: Model 2 uses Monte Carlo cross-validation to calculate the f1-score, ensuring a reliable assessment of its performance.

However, the added complexity of the exhaustive search methodology comes at a computational cost, making Model 2 orders of magnitude slower than Model 1. Despite this, Model 2 remains relatively quick, with computation times still measured in seconds. Overall, Model 2 offers a more comprehensive and robust approach to LDA, making it a suitable choice for applications where feature selection and handling collinearity are crucial.

Model 3 &4: Enhanced QDA with Exhaustive Search and Feature Selection

Model 4 builds upon the foundation of Quadratic Discriminant Analysis (QDA), which uses a quadratic surface to separate classes. This model enhances the basic QDA approach (Model 3) by incorporating an exhaustive search methodology for feature selection, similar to the approach used in Model 2.

Key characteristics of Model 4 include:

- Quadratic separation: Model 4 utilizes a quadratic surface to distinguish between classes, allowing for more complex relationships between features to be captured.
- Normal distribution requirement: Features must follow a normal distribution for the model to be effective.
- Varying covariances: Unlike some other models, QDA in Model 4 allows covariances to vary for each class, making it more flexible in handling diverse data structures.
- No scaling requirement: Scaling of features is not necessary for Model 4, simplifying the preprocessing steps.
- Monte Carlo cross-validation: This method is used to calculate the f1-score, providing a robust evaluation of the model's performance.

The exhaustive search methodology in Model 4 generates 2^p models (where p is the number of features) and selects the model with the best combination of features that yields the highest f1-score. This approach:

1. Addresses variable selection: By evaluating all possible combinations of features, Model 4 can better handle variable selection, although it still does not inherently address collinearity issues.
2. Improves over basic QDA: The exhaustive search enhances the model's ability to find the most relevant features, potentially leading to better performance than a basic QDA model that uses all features without selection.

While Model 4 offers these advantages, its computational time is longer than the basic QDA model due to the exhaustive search, but it remains relatively quick, with computation times still measured in seconds. Overall, Model 4 provides a more comprehensive approach to QDA, suitable for applications where capturing complex relationships between features and selecting the most relevant variables are crucial.

Model 5 & 6: Enhanced Naive Bayes (NB) with Exhaustive Search and Feature Selection

Model 6 builds upon the foundation of Naive Bayes (NB), a probabilistic model that determines the probability of an outcome given a set of conditions using Bayes' theorem. This model enhances the basic NB approach by incorporating an exhaustive search methodology for feature selection.

Key characteristics of Model 6 include:

- Bayes' theorem application: Model 6 uses Bayes' theorem to calculate the probability of an outcome based on given conditions, providing a robust framework for probabilistic predictions.
- Flexible feature distributions: Unlike some other models, NB in Model 6 can handle features that follow any distribution, making it versatile for various data types.
- Varying covariances: Model 6 allows covariances to vary for each class, which is beneficial for datasets where the relationship between features differs significantly across classes.
- No scaling requirement: Scaling of features is not necessary for Model 6, simplifying the preprocessing steps.
- Monte Carlo cross-validation: This method is used to calculate the f1-score, providing a reliable evaluation of the model's performance.

The exhaustive search methodology in Model 6 generates 2^p models (where p is the number of features) and selects the model with the best combination of features that yields the highest f1-score. This approach:

1. Enhances feature selection: By evaluating all possible combinations of features, Model 6 can identify the most relevant features for the model, potentially improving its performance.
2. Improves over basic NB: The exhaustive search in Model 6 addresses one of the limitations of basic NB models, which assume independence between features. While NB itself does not account for interactions between features, the feature selection process can help mitigate this issue by selecting features that are most informative when considered together.

Model 6 retains the computational efficiency of the basic NB model, with computation times still measured in seconds, despite the added complexity of the exhaustive search. This makes Model 6 a suitable choice for applications where rapid, accurate classification is necessary, and the relationships between features are complex. Overall, Model 6 offers a more comprehensive and robust approach to NB, combining the strengths of probabilistic modeling with thorough feature selection.

Model 7,8 &9: KNN, KNN with Exhaustive Search, and PCA then KNN

K-Nearest Neighbors (KNN) is a simple, non-parametric machine learning algorithm used for classification tasks. It predicts the label of a data point by looking at the k closest data points in the dataset. The theory is that the label class of particular data point will be the average label of its neighbors. Before use, the k closest data points (number of closest neighbors) have to be optimized. It's easy to implement but can be slow and sensitive to noisy or high-dimensional data.

Model 8 is just exhaustive search performed on model 7 as described in model 2 and 4 above.

Model 9 represents a comprehensive approach to K-Nearest Neighbors (KNN) modeling, incorporating both Principal Component Analysis (PCA) for dimensionality reduction. This model builds upon the foundations of Model 7, which employed KNN with all features.

Key Components and Methodology:

- **PCA for Dimensionality Reduction:** Initially, PCA is applied to minimize the dimensionality of the features. As illustrated in [Figure 6](#), the first 6 principal components capture approximately 90% of the variance in the dataset, indicating that these components are sufficient for representing the underlying structure of the data.
- **KNN Algorithm:** Following PCA, a KNN algorithm is implemented on the reduced dataset with 6 components. This approach enables the model to leverage the strengths of KNN in averaging the target class values near the point of interest, while also benefiting from the dimensionality reduction achieved through PCA.

Optimization and Performance

The optimal number of neighbors for the KNN algorithm is determined to be 15 for in the case of model 7, as shown in [Fig. 5](#). For models 8 and 9, obviously, the optimal number of neighbors will change depending on the number of features selected.

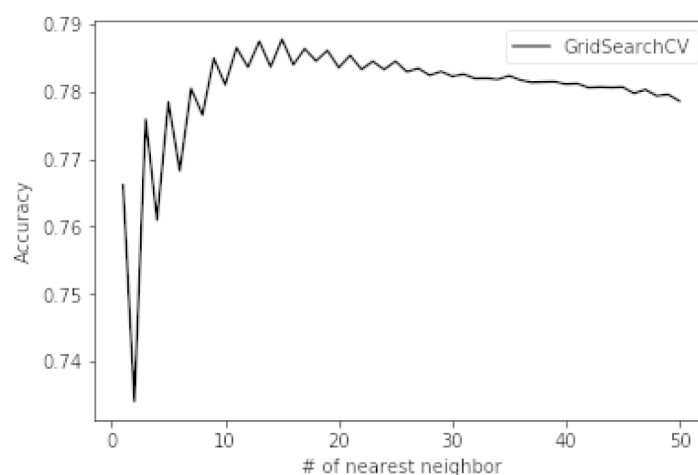


Figure 5—Cross-validation to select the optimal numbers of neighbors

Challenges and Considerations

One of the challenges associated with this approach is translating the principal components from the PCA into physical terms, as they encompass both geological and engineering aspects. However, the benefits of using PCA with KNN can outweigh these challenges, making Model 9 a reliable choice for classification tasks.

Computational Efficiency and Practical Applications

While the exhaustive search process in Model 8 is computationally demanding, the overall model retains a reasonable level of efficiency. The dimensionality reduction achieved through PCA helps to mitigate the computational costs associated with the exhaustive search, making Model 9 a viable option for applications where the interpretation of the principal components are relatively straightforward. Ultimately, Model 9 can be a powerful and refined approach to KNN modeling, leveraging the strengths of PCA, and KNN to deliver high-performance classification capabilities.

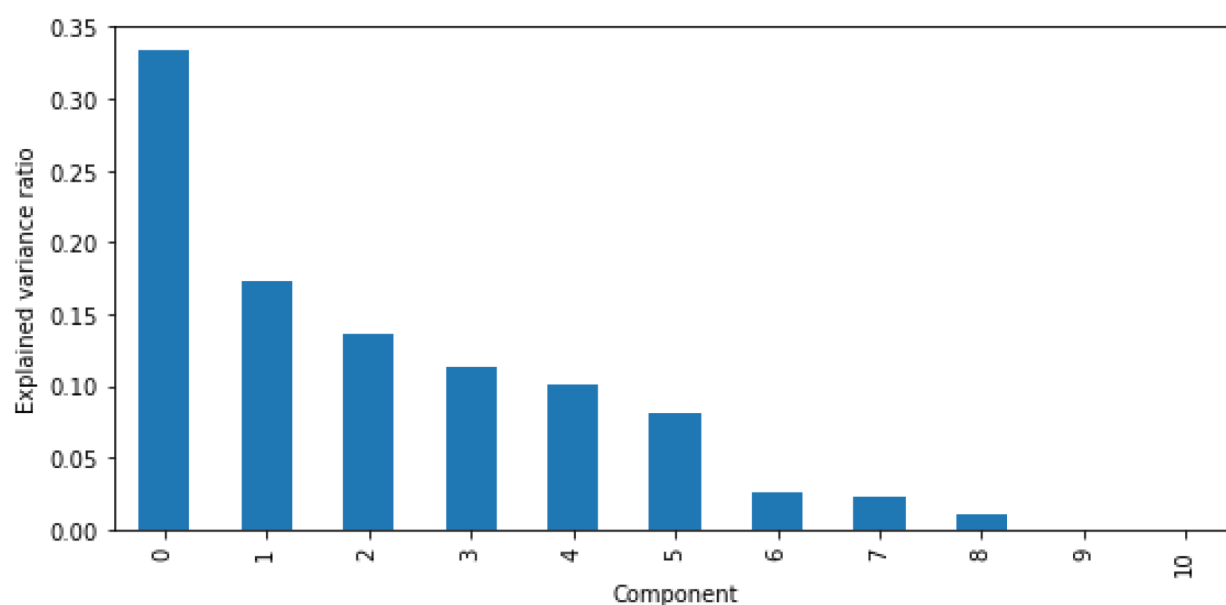


Figure 6—PCA use that first 6 features explain #90% of the variance

Model 10 & 11: Optimized Random Forest Classifier

Model 11 represents an enhanced Random Forest classifier, building upon the foundation of Model 10, which utilized the default hyperparameters of the Random Forest algorithm. In Model 11, the hyperparameters are carefully tuned to achieve a higher score, leveraging the strengths of the Random Forest approach, which combines the predictions of multiple decision trees to produce a more accurate classification. Table 4 shows the parameters and the corresponding tuned value.

Table 4—Hyper parameters for Random forest.

Parameter	Description
N_Estimators	Number of trees in the forest
Criterion	What criterion is used to split a node (make a branch)
Max_Depth	The maximum depth of the tree
Max_features	The number of features to consider when looking for the best split
Min_Samples_Leaf	The minimum number of samples required to be at a leaf node
Min_Samples_Split	Minimum number of samples required to split an internal node

Optimization Process

The optimization of Model 11 involves a systematic approach to tuning the hyperparameters. This process allows for the identification of the most critical hyperparameters and their optimal ranges. A "Gridsearch" algorithm is then employed to exhaustively search for the optimal combination of hyperparameter values, resulting in the highest f1-score. The optimization of the min sample leaf is shown in Figure 7 as an example.

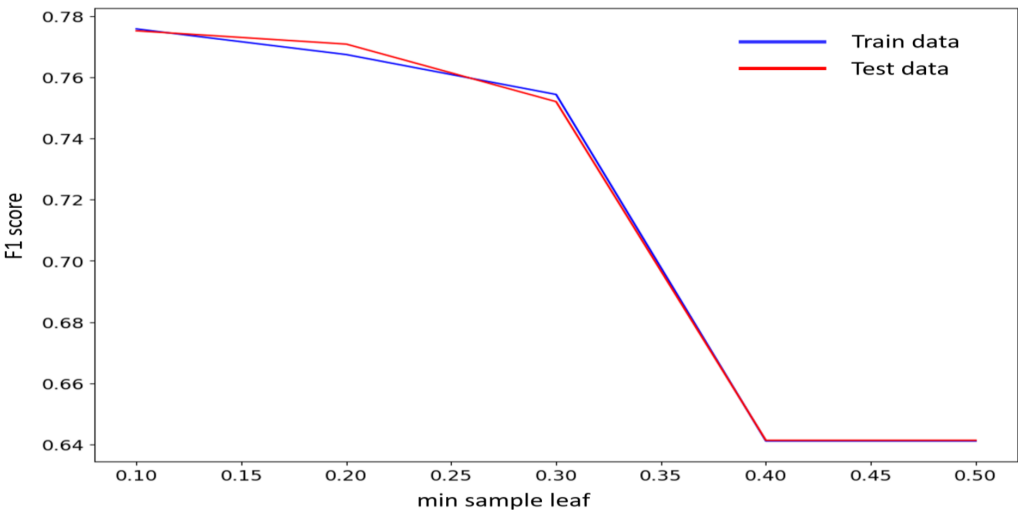


Figure 7—Min sample leaf vs f1-score show that the best f1 score is when the min leaf is 0.1. This helps narrow down the search space of the Gridsearch algorithm.

Hyperparameter Tuning

Table 4 provides a detailed overview of the hyperparameters, while table 5 gives their final tuned values. The tuning process involves adjusting the hyperparameters to achieve the best possible performance.

Table 5—Final tuned value for random forest

Parameter	Final Tuned Values
N_Estimators	190
Criterion	Entropy
Max_Depth	14
Max_Features	None
Min_Samples_Leaf	0.1
Min_Samples_Split	0.1

Improved Performance

The optimized hyperparameters in Model 11 lead to an improvement in the model's performance, as evidenced by the increased f1-score. The careful tuning of the hyperparameters ensures that the Random Forest classifier is able to effectively capture the underlying patterns in the data, resulting in more accurate classifications. Overall, Model 11 represents a robust and optimized Random Forest classifier, capable of delivering high-performance results in classification tasks. The final tuned values is in table 5.

Model 12 & 13: Optimized Gradient Boosting

Model 13 represents an advanced Gradient Boosting approach, building upon the foundation of Model 12. Gradient Boosting is a powerful ensemble learning technique that shares similarities with Random Forest, but with a key difference: each subsequent tree is trained to minimize the prediction error of the previous tree. This autoregressive nature enables Gradient Boosting to be faster and more accurate than Random Forest, but it also requires more hyperparameter tuning.

Key Characteristics and Hyperparameter Tuning

The Gradient Boosting algorithm used in Model 13 is characterized by its ability to iteratively improve the predictions by fitting the residuals of the previous model. This process continues until the desired level of accuracy is achieved. The hyperparameters used to tune the Gradient Boosting model are similar to those used in Random Forest, but with some notable differences. The final tuned values for the Gradient Boosting model are presented in Table 6

Table 6—GB Final tuned values

Parameter	Final Tuned Values
N_Estimators	210
Loss	Deviance
Max_Depth	None
Max_Features	None
Min_Samples_Leaf	0.1
Min_Samples_Split	0.1
Criterion	Friedman MSE
Learning Rate	0.2

Optimization and Performance

The optimized hyperparameters in Model 13 result in improvement in the model's performance, with increased accuracy and efficiency. The Gradient Boosting algorithm's ability to iteratively improve the predictions by minimizing the prediction error of the previous tree enables it to capture complex patterns in the data. The careful tuning of the hyperparameters ensures that the model is able to effectively balance the trade-off between bias and variance, resulting in a robust and accurate classifier. Overall, Model 13 represents a state-of-the-art Gradient Boosting approach, capable of delivering high-performance results in classification tasks.

Results and Model Comparison

The results of the model evaluation are presented in Table 7, which shows the f1-score and variables used to calculate it for each model. Based on the results, several key observations can be made:

Table 7—Summary of all models along with the Score and features.s

Model #		Geological Variables						Engineering Variables					F1-Score
		Lat	Long	Gravity (heavy)	Gravity (light)	Gravity (condensate)	Gravity (Ultra Light)	Prop Vol	Fluid Vol	TVD	LL	Avg. well spacing	
1	LDA (All feat)	x	x	x	x	x	x	x	x	x	x	x	0.83
2	LDA (Ex. Search)					x		x	x		x		0.84
3	QDA (All feat)	x	x	x	x	x	x	x	x	x	x	x	0.76
4	QDA(Ex. Search)				x		x	x			x		0.84
5	NB (All feat)	x	x	x	x	x	x	x	x	x	x	x	0.8
6	NB (Ex. Search)				x		x			x	x		0.84
7	KNN (All feat)	x	x	x	x	x	x	x	x	x	x	x	0.84
8	KNN ((Ex. Search)	x	x					x		x			0.86
10	R.F (Untuned)	x	x					x	x	x	x	x	0.88
11	R.F (Tuned)	x	x					x		x	x	x	0.9
12	G.B (Untuned)	x	x					x	x	x	x	x	0.88
13	G.B (Tuned)	x	x					x	x	x	x	x	0.89

		PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	F1-Score
9	PCA then KNN	x	x	x	x	x							0.83

1. Model 11: The Best Performing Model: Model 11, the optimized Random Forest classifier, emerges as the top-performing model, achieving the highest f1-score. This is attributed to Random Forest's exceptional ability to handle high-dimensional data, its speed, and its capacity to manage outliers and non-linear relationships. As evident in Figure 4, the data exhibits non-linear characteristics and

contains several outliers, making Random Forest particularly well-suited for this task. In terms of variable selection, Model 11 chose all geological features (latitude, longitude) except gravity, and all engineering variables except fluid vol. Notably, [Figure 4](#) reveals a strong correlation between prop volume and fluid volume, which may have influenced the model's variable selection.

2. **LDA Outperforms QDA:** Contrary to expectations, LDA outperformed QDA. QDA is typically better suited for non-linear data, but in this case, LDA's performance was superior. This unexpected result warrants further investigation to understand the underlying factors contributing to LDA's better performance.
3. **Random Forest Surpasses Gradient Boost:** Another unexpected result is the superior performance of Random Forest over Gradient Boost. Given Gradient Boost's ability to handle complex interactions and non-linear relationships, it was anticipated to outperform Random Forest. However, the numerous hyperparameters utilized in Gradient Boost may have contributed to its relatively poorer performance. The complexity of tuning these hyperparameters may have led to suboptimal performance, highlighting the importance of careful hyperparameter tuning in Gradient Boost models.

These findings suggest that Model 11, the optimized Random Forest classifier, is the most suitable choice for this classification task, due to its exceptional performance and ability to handle the complexities of the data. The unexpected results for LDA and QDA, as well as the comparison between Random Forest and Gradient Boost, underscore the importance of thorough model evaluation and hyperparameter tuning to achieve optimal performance.

Prescriptive Statistics.

Following the application of the best classification model to the dataset, the next step involves exploring the feasibility of transforming wells with a negative Net Present Value (NPV) into profitable ones. The analysis focuses on the 1653 wells with a negative NPV, as shown in [Figure 8](#).

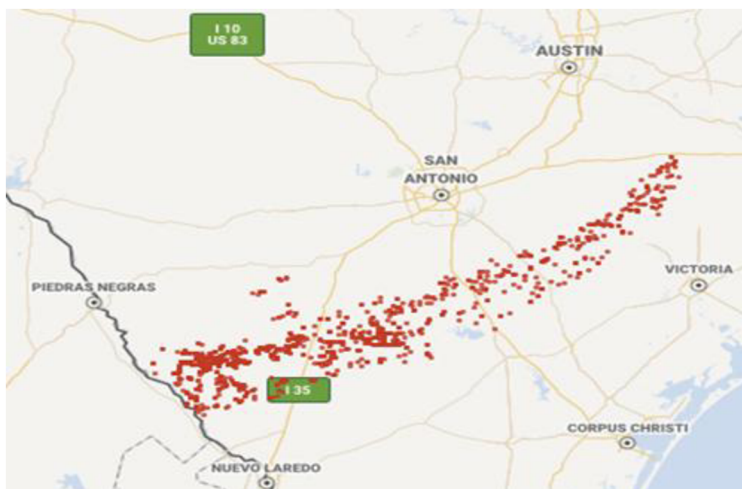


Figure 8—Total number of wells classified to have negative NPV

Engineering Variable Optimization

Since geological variables are immutable, the study concentrates on the impact of modifying engineering variables on wells with negative NPV. The engineering dataset for these wells is adjusted to different percentiles of the engineering dataset with a positive NPV. The model is then re-run, and the fraction of "bad" wells that are converted to "good" wells is calculated in [Figure 9](#).

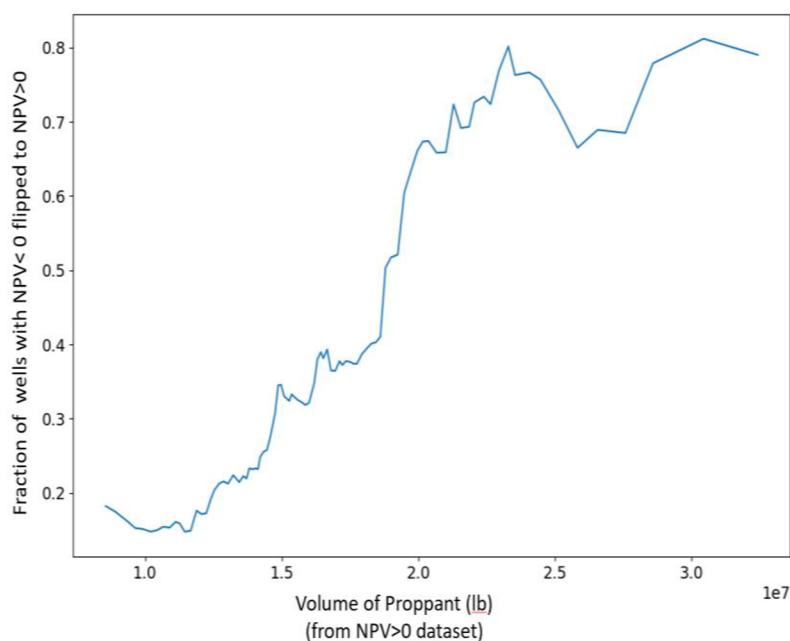


Figure 9—Plot of Volume of proppant and fraction of bad wells turned good

Figures 9 to 11 result of the analysis reveal a point of diminishing returns for engineering efforts, where further investments in certain variables yield minimal improvements. Notably, the optimal values for these variables are approximately 24 million pounds of sand for the volume of proppant, 9,500 ft for the lateral length, and 900 ft for the average well spacing. Furthermore, the analysis highlights that the majority of the improvement in well performance occurs within specific ranges of these variables, including 18-20 million pounds for the volume of proppant, 8,000-8,800 ft for the lateral length, and 300-800 ft for the well spacing. These findings suggest that engineering efforts should focus on optimizing these variables within these critical ranges to maximize returns and minimize unnecessary investments.

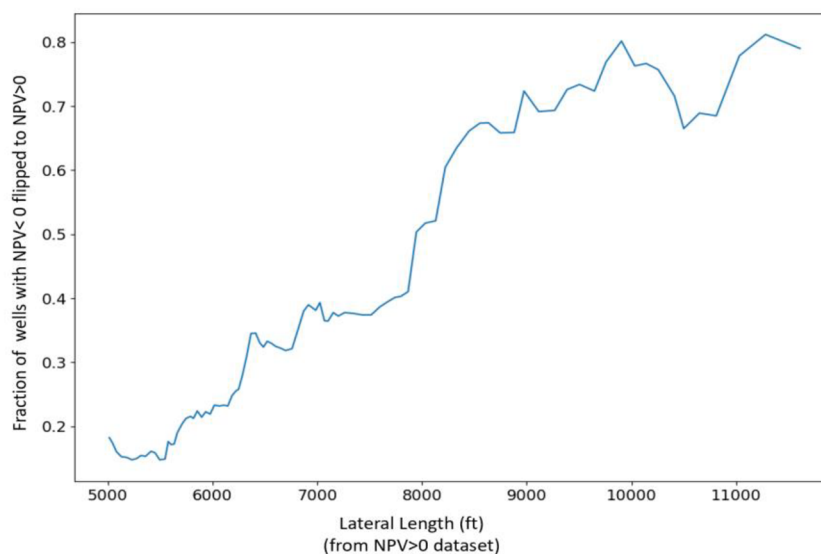


Figure 10—Plot of Lateral Length and fraction of bad wells turned good

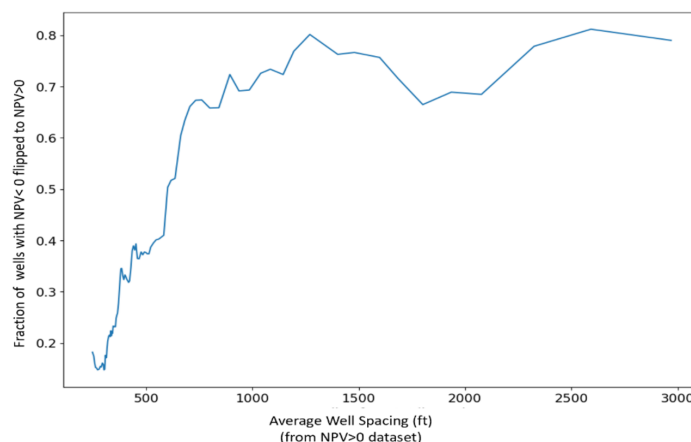


Figure 11—Plot of Avg Well Spacing and fraction of bad wells turned good

Spatial Distribution of Upgraded Wells

Figures 12 to 14 illustrate the number of wells that were upgraded from class 0 to class 1 as the percentile of the engineering variables was changed. The results indicate that there is still a strong location-dependent component in the number of wells that remained unchanged. The spatial distribution of the original "bad" wells (red) that turned into "good" wells at different engineering variable percentiles is also presented.

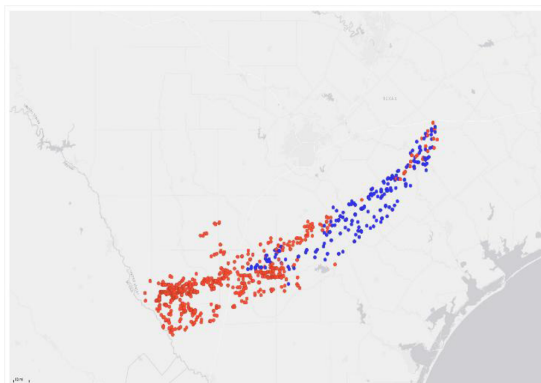


Figure 12—Number of wells that were upgraded from class 0 to 1 (blue), and number of wells still in class 0 at a 25-percentile engineering variable. Class 0 is red, Class 1 in blue

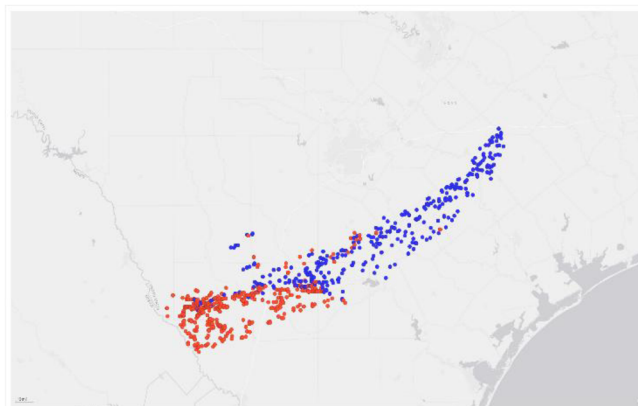


Figure 13—Number of wells that were upgraded from class 0 to 1 (blue), and number of wells still in class 0 at a 50-percentile engineering variable

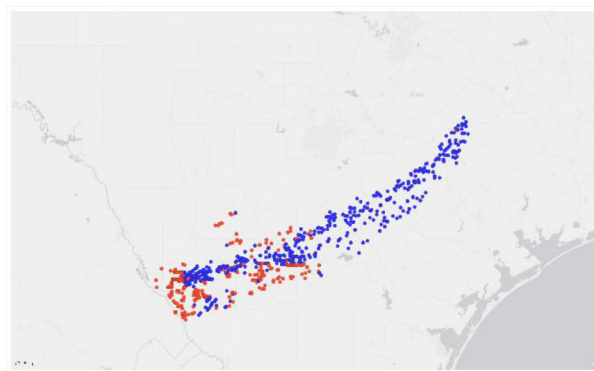


Figure 14—Number of wells that were upgraded from class 0 to 1 (blue), and number of wells still in class 0 at a 75-percentile engineering variable. Class 0 is red, Class 1 in blue

Conclusions and Recommendations

This study applied 13 different classification models to distinguish between wells with negative Net Present Value (NPV) and those with positive NPV. After evaluating the performance of each model, the Random Forest classifier emerged as the best model, based on its superior f1-score. The most influential features contributing to the model's performance were identified as latitude, longitude, prop vol, TVD, lateral length, and average well spacing.

The analysis also explored the impact of modifying engineering variables on the classification of wells with negative NPV. By adjusting the engineering variables from NPV less than zero to different percentiles of the engineering variables, the number of "bad" wells that flipped to "good" wells changed significantly. Furthermore, the study revealed a strong spatial dependency in the classification model, indicating that the likelihood of successfully engineering a well to improve its NPV is highly dependent on its location within the basin.

Notably, the results suggest that it is highly unlikely to engineer wells in the southern west part of the basin to improve their NPV, implying that resources may be better allocated to other areas with more promising potential. These findings have important implications for operators and investors seeking to optimize their well development strategies and maximize returns on investment.

Based on the conclusions of this study, the following recommendations are made:

1. Focus on high-potential areas: Prioritize well development in areas with favorable geological and engineering conditions, as identified by the classification model.
2. Optimize engineering variables: Adjust engineering variables, such as prop vol, lateral length, and average well spacing, to optimize well performance and improve NPV.
3. Avoid low-potential areas: Be cautious when considering well development in areas with low potential, such as the southern west part of the basin, where engineering efforts are unlikely to yield significant improvements.
4. Continuously monitor and evaluate: Regularly monitor and evaluate well performance, and adjust development strategies accordingly, to ensure optimal resource allocation and maximize returns on investment.

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